The Log-Discriminant Hessian for Near-Boundary Quadrature

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Abstract

We study a polynomial-algebraic discretization of the hypersingular boundary operator that controls near-boundary layer-potential quadrature. For distinct points $V = (v_1, \dots, v_n) \subset \mathbb{C}$ let

$$\Delta(V) = \prod_{i<j} (v_i - v_j)^2,$$
$$Q(V) = -\tfrac{1}{2} \nabla_V^2 \log |\Delta(V)|.$$

In complex coordinates this Hessian is the chord-squared graph Laplacian

$$Q_{ij} = -|v_i - v_j|^{-2} \quad (i \neq j),$$
$$Q_{ii} = \sum_{k \neq i} |v_i - v_k|^{-2}.$$

On the regular n-gon it is circulant and has the exact spectrum $\lambda_m = m(n-m)/2$, $m = 0, \dots, n-1$. After multiplication by the arclength weight this spectrum converges to that of $\pi(-\Delta_{S^1})^{1/2}$, and on a C^3 closed curve the continuum operator

$$(Q_\Gamma f)(s) = \text{p. v.} \int_\Gamma \frac{f(s) - f(t)}{|\gamma(s) - \gamma(t)|^2} \, dt$$

satisfies $Q_\Gamma = \pi \Lambda_\Gamma + K_0$, where Λ_Γ is the Dirichlet-to-Neumann map and $K_0 \in \Psi^0(\Gamma)$. This gives an intrinsic explanation of the $n\delta \sim 1$ breakdown threshold for the periodic trapezoidal rule near a boundary: the error splits into a bulk analytic-continuation channel and a local channel with universal profile $\operatorname{Re} \log(1 - e^{-2\pi\rho + 2\pi i \beta})$, $\rho = \delta/h$. The same operator yields an $O(n \log n)$ functional calculus on the circle, an $O(n)$-conditioned hypersingular discretization modulo lower order terms, and a minimax quadrature rate on analytic Jordan curves governed by the sum of the density collar modulus and the target conformal distance.

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1 Introduction

Layer potentials give high-order solvers for planar boundary value problems, but their direct quadrature becomes delicate when the target approaches the source curve. For a smooth closed curve $\Gamma \subset \mathbb{C}$ and a density σ consider

$$I(x) = \int_\Gamma \log |x - y| \, \sigma(y) \, ds(y), \qquad x \notin \Gamma. \quad (1)$$

The periodic trapezoidal rule is exponentially accurate when the target is separated from Γ by a distance large compared with the mesh width [13]. As soon as the normal distance $\delta = \operatorname{dist}(x, \Gamma)$ becomes comparable with the spacing $h = L/n$, the logarithmic kernel has a singularity within one mesh cell of the contour and the error no longer follows the resolved analytic rate. This is the near-singular quadrature problem addressed by specialized methods such as panel refinement, singularity subtraction, and quadrature by expansion [3, 6, 5].

There are two scales in this failure. The first is analytic: the classical trapezoidal estimate is controlled by how far the integrand can be continued off the real parameter axis before meeting a singularity. The second is discrete and local: when the nearest complex singularity lies within $O(h)$ of a node, the error is set by the unresolved tangent-line logarithm rather than by the global analytic strip. The numerical literature handles the second scale very effectively by changing the quadrature rule. The purpose here is different: identify the boundary operator whose spectrum and symbol explain why the second scale is intrinsic.

This paper isolates a single operator behind that transition. It is not introduced by regularizing the singular integral. It is the Hessian of the logarithm of a polynomial discriminant. The resulting matrix is a weighted graph Laplacian with weights $|v_i - v_j|^{-2}$, and on roots of unity it has a closed form Fourier spectrum. The continuum limit is the principal part of the hypersingular boundary integral operator, equivalently the Dirichlet-to-Neumann map up to the factor π and a bounded zeroth-order remainder.

The main contributions are as follows.

1. The discrete log-discriminant Hessian Q_n on the regular n-gon has eigenvalues $m(n-m)/2$. Its weighted continuum scaling has symbol $\pi|\xi|$, giving an FFT functional calculus at $O(n \log n)$ cost.

2. On a C^3 curve, the continuum inverse-square operator Q_Γ equals $\pi\Lambda_\Gamma$ modulo Ψ^0. Therefore the chord-squared discretization is a direct Nyström discretization of the hypersingular operator's order-one part.

3. The trapezoidal error near Γ has a two-channel decomposition: a bulk channel controlled by analytic continuation and a local channel controlled by the universal profile

$$\mathcal{M}(\rho, \beta) = \operatorname{Re} \log(1 - e^{-2\pi\rho + 2\pi i\beta}), \qquad \rho = \delta/h.$$

   The crossover occurs at $\rho = O(1)$, i.e., $n\delta/L = O(1)$.

4. On analytic Jordan curves the minimax error over Hardy-class densities has exponential rate

$$n^{-1} \exp\Big[-\frac{n}{2}\big(\log\rho + \delta_\Phi(x)\big)\Big],$$

   where $\log\rho$ is the density collar width and $\delta_\Phi(x)$ is the exterior conformal distance of the target.

Longer geometric material from the source manuscript, including the discriminant-cone interpretation and inverse-spectral pipeline, is not used in the proofs below and is omitted from this SIAM-length version.

The organization follows the chain from algebra to numerics. Section 2 derives the log-discriminant Hessian and its exact polygon spectrum, including the spectral-renormalization identity that explains why logarithmic coordinates and zeta tails appear. Section 3 passes to smooth curves and identifies the continuum operator with the Dirichlet-to-Neumann map modulo lower order terms. Section 4 proves the near-singular bridge and separates the bulk and local error channels. Section 5 states the analytic minimax rate in conformal coordinates. Section 6 gives the reduced numerical evidence, and Section 7 records the geometric reading and limitations.

2 The Algebraic Operator

Let $V = (v_1, \dots, v_n)$ be a configuration of distinct complex points. The discriminant

$$\Delta(V) = \prod_{1 \le i < j \le n} (v_i - v_j)^2$$

vanishes precisely on collisions. On the collision-free configuration space the function

$$\Phi(V) = \log|\Delta(V)| = 2\sum_{i<j} \log|v_i - v_j|$$

is real analytic. Its negative half Hessian is the matrix

$$Q_{ij}(V) = \begin{cases} -|v_i - v_j|^{-2}, & i \neq j, \\ \displaystyle\sum_{k \neq i} |v_i - v_k|^{-2}, & i = j. \end{cases} \tag{2}$$

The identity follows by differentiating each two-point logarithm in complex coordinates. The row sums vanish and the quadratic form is

$$u^* Q(V) u = \frac{1}{2} \sum_{i \neq j} \frac{|u_i - u_j|^2}{|v_i - v_j|^2}. \tag{3}$$

Thus $Q(V)$ is positive semidefinite and has kernel exactly the constant vector when the interaction graph is connected.

2.1 Log-Hessian identity

The definition is coordinate-free enough for the present purposes, but the elementary calculation is worth recording because it fixes the normalization. Write a complex point as $v = x + iy$. For $i \neq j$,

$$\nabla_{v_i} \nabla_{v_j} \log|v_i - v_j|$$

has the real 2×2 block of the Coulomb Hessian. When the normal/tangential degrees of freedom are compressed onto scalar boundary data, the mixed scalar block is $-|v_i - v_j|^{-2}$. Summing the pair interactions over all unordered pairs gives the diagonal completion in (2). Equivalently, for scalar variations $u = (u_1, \dots, u_n)$,

$$-\frac{1}{2}\frac{d^2}{d\varepsilon^2}\Phi(v_1+\varepsilon u_1,\dots,v_n+\varepsilon u_n)\Big|_{\varepsilon=0} = \frac{1}{2}\sum_{i\neq j}\frac{|u_i-u_j|^2}{|v_i-v_j|^2},$$

after taking the boundary scalar component of the variation. This is why the Hessian is a Laplacian rather than a general dense symmetric matrix.

Three invariances are immediate. Translation of all v_i leaves $Q(V)$ unchanged. Rotation and dilation by $a \in \mathbb{C}^\times$ multiply the weights by $|a|^{-2}$. Relabeling nodes conjugates the matrix by the corresponding permutation. These invariances are exactly the similarity invariances expected of the principal part of a boundary operator.

2.2 The regular polygon

Set $\omega = e^{2\pi i/n}$ and $V_\omega = (1, \omega, \dots, \omega^{n-1})$. The corresponding matrix $Q_n := Q(V_\omega)$ is circulant. Its first row is

$$(Q_n)_{00} = \sum_{k=1}^{n-1}\frac{1}{|1-\omega^k|^2},$$
$$(Q_n)_{0k} = -\frac{1}{|1-\omega^k|^2} \quad (k \neq 0).$$

Theorem 2.1 (polygon spectrum)**.** *For $m = 0, \dots, n-1$, the Fourier mode $\varphi_m(j) = n^{-1/2}\omega^{mj}$ is an eigenvector of Q_n with*

$$\lambda_m(Q_n) = \frac{m(n-m)}{2}. \tag{4}$$

Consequently $\ker Q_n = \operatorname{span}\{\mathbf{1}\}$ *and* $\operatorname{cond}(Q_n|_{\mathbf{1}^\perp}) = O(n)$.

Proof. For a circulant graph Laplacian the eigenvalue on mode m is

$$\lambda_m = \sum_{k=1}^{n-1}\frac{1-\omega^{mk}}{|1-\omega^k|^2}.$$

The imaginary part cancels by pairing k and $n-k$. The real part is

$$\frac{1}{2}\sum_{k=1}^{n-1}\frac{2-\omega^{mk}-\omega^{-mk}}{(1-\omega^k)(1-\omega^{-k})}.$$

Using

$$\frac{1-z^m}{1-z} = 1 + z + \dots + z^{m-1}, \qquad z = \omega^k,$$

the summand becomes one half of a finite geometric convolution. Orthogonality of the nontrivial roots of unity leaves exactly the $m(n-m)$ surviving pairs. This gives (4). ☐

The arclength-weighted matrix

$$\mathsf{Q}_n = \frac{2\pi}{n}Q_n \tag{5}$$

has eigenvalues $\pi m(1-m/n)$ for $0 \le m \le n-1$. For fixed Fourier index m and $n \to \infty$ this converges to $\pi|m|$, the spectrum of $\pi(-\Delta_{S^1})^{1/2}$.

Corollary 2.2 (functional calculus)**.** *Let φ be defined on* $\operatorname{spec}(Q_n)$*. If F_n is the unitary DFT, then*

$$\varphi(Q_n)u = F_n^* \operatorname{diag}\big(\varphi(m(n-m)/2)\big)_{m=0}^{n-1}F_n u.$$

Each application costs $O(n\log n)$. On $\mathbf{1}^\perp$ this gives fast inverses, fractional powers, heat semigroups, and spectral filters.

Remark 2.3. The null flux identity $Q_n\mathbf{1} = 0$ is algebraic. It is not a quadrature accident and does not depend on cancellation at machine precision. This matters for hypersingular equations, where failure to preserve the constant nullspace produces visible flux leakage.

2.3 Cauchy–Gram factorization

The spectrum in Theorem 2.1 can also be read as a Cauchy–Gram identity. Let $p(z) = z^n - 1$ and write

$$\frac{p'(z)}{p(z)} = \sum_{j=0}^{n-1}\frac{1}{z-\omega^j}.$$

Taking boundary traces on the roots and subtracting the singular self-term produces the same inverse chord weights as (2). In Fourier variables, multiplication by z^m and the root identities

$$\sum_{j=0}^{n-1}\omega^{j\ell} = \begin{cases} n, & \ell \equiv 0 \pmod n, \\ 0, & \text{otherwise} \end{cases}$$

leave only the pairs (a,b) with $0 \le a < m$ and $m \le b < n$. Their number is $m(n-m)$, and the factor $1/2$ is the symmetrization between the two endpoints of the chord. This is the finite-dimensional form of the Cauchy projection identity behind the half-Laplacian on the circle.

2.4 Spectral renormalization

The quadratic spectrum $m(n-m)/2$ contains a bulk term and two boundary tails. This decomposition is useful because it is the same bulk/local splitting that reappears in the quadrature bridge. Define

$$S_n(s) = \sum_{m=1}^{n-1} \left(\frac{m(n-m)}{n} \right)^{-s}. \tag{6}$$

For $\operatorname{Re} s < 1$ the corresponding bulk integral is $n^{1-s}B(1-s, 1-s)$; both sides then extend meromorphically.

Theorem 2.4 (renormalized spectral tail)**.** *Let*

$$T_n(s) = S_n(s) - n^{1-s}B(1-s, 1-s).$$

For every compact $K \subset \mathbb{C} \setminus \mathbb{Z}_{\geq 1}$ there is C_K such that

$$|T_n(s) - 2\zeta(s)| \leq C_K n^{-1}, \qquad s \in K, \quad n \geq 2. \tag{7}$$

In particular $T_n(0) = -1 = 2\zeta(0)$.

Proof sketch. Apply Euler–Maclaurin to $f_s(x) = (x(1-x))^{-s}$ on $[1/n, 1 - 1/n]$ after cutting off the two endpoints. The interior contribution is $n^{1-s}B(1-s, 1-s)$, while the two endpoint tails are

$$\sum_{m \geq 1} m^{-s} + \sum_{m \geq 1} m^{-s} = 2\zeta(s)$$

after analytic continuation. The matching terms at the cutoffs cancel between the bulk integral and the truncated tails. Uniformity on compact subsets away from the positive integers follows either from the Euler–Maclaurin remainder with a fixed number of endpoint subtractions or from the Mellin–Barnes contour representation of $(1-x)^{-s}$. The direct identity at $s = 0$ is $(n-1) - n = -1$. □

The theorem says that the zeta function is not an extra analytic object added to the polynomial calculation; it is the residual boundary tail of the finite discriminant spectrum after the unique bulk counterterm has been removed. This is also why logarithms rather than powers are the natural coordinates in the near-boundary estimates: multiplicative collars become additive distances after the Mellin renormalization.

3 Continuum Limit and Hypersingular Operators

Let Γ be a closed C^3 Jordan curve parametrized by arclength $\gamma : \mathbb{R}/L\mathbb{Z} \to \mathbb{C}$. Define

$$(Q_\Gamma f)(s) = \text{p. v.} \int_0^L \frac{f(s) - f(t)}{|\gamma(s) - \gamma(t)|^2} \, dt. \tag{8}$$

The principal singularity is the one-dimensional inverse-square kernel. Since $|\gamma(s) - \gamma(t)|^2 = (s-t)^2 + O(|s-t|^4)$, the local model is

$$\text{p. v.} \int_{\mathbb{R}} \frac{f(s) - f(t)}{(s-t)^2} \, dt = \pi |D| f(s).$$

The curvature-dependent difference is less singular.

Lemma 3.1 (tangent-graph decomposition)**.** *In every sufficiently small arclength chart,*

$$\frac{1}{|\gamma(s) - \gamma(t)|^2} = \frac{1}{(s-t)^2} + b_0(s,t), \tag{9}$$

where b_0 is locally integrable as a finite-part kernel and defines a pseudodifferential contribution of order 0. More precisely, if $\gamma \in C^3$, then after subtracting the tangent-line kernel the remaining operator maps $H^{1/2}$ to $H^{1/2}$ locally.

Proof. Taylor expansion in arclength gives

$$\gamma(t) = \gamma(s) + (t-s)T(s) + \tfrac{1}{2}(t-s)^2 \kappa(s) N(s) + O(|t-s|^3).$$

Because $T \cdot N = 0$ and $|T| = 1$,

$$|\gamma(t) - \gamma(s)|^2 = (t-s)^2 - \frac{\kappa(s)^2}{12}(t-s)^4 + O(|t-s|^5).$$

Therefore the difference between the reciprocal chord square and $(s-t)^{-2}$ is bounded at the diagonal up to the finite smoothness allowed by γ. The off-diagonal part is smooth, so the induced operator has order zero. □

Theorem 3.2 (Dirichlet-to-Neumann identification)**.** *On mean-zero functions on a C^3 closed curve,*

$$Q_\Gamma = \pi \Lambda_\Gamma + K_0, \tag{10}$$

where Λ_Γ is the interior Dirichlet-to-Neumann map and $K_0 \in \Psi^0(\Gamma)$. In particular Q_Γ is an elliptic order-one pseudodifferential operator with principal symbol $\pi|\xi|$.

Proof. Work in an arclength chart near $s = t$. Taylor expansion gives

$$\frac{1}{|\gamma(s) - \gamma(t)|^2} = \frac{1}{(s-t)^2} + a_0(s,t),$$

where a_0 is a conormal kernel of order 0 under the stated regularity. The first term has Fourier multiplier $\pi|\xi|$. The Dirichlet-to-Neumann map has principal symbol $|\xi|$ [4, 8]. Patching the local charts by a partition of unity leaves a global zeroth-order remainder. □

Remark 3.3 (what the theorem does and does not say). The identity (10) is a principal-symbol statement with a controlled lower-order remainder. It does not claim that the chord kernel is a new formula for the full Dirichlet-to-Neumann map on an arbitrary curve. It says that the algebraic Hessian selects the canonical order-one part of that map without choosing a regularization of a hypersingular integral.

Proposition 3.4 (rigidity on the circle)**.** *On S^1, suppose A is a real, translation-invariant, positive-semidefinite operator on mean-zero trigonometric polynomials whose quadratic form is invariant under rotations, has order-one homogeneity under dilation of the boundary length, and agrees with the two-point inverse-square interaction on localized differences. Then $A = c|D|$ for some $c > 0$. With the normalization of* (8)*, $c = \pi$.*

Proof. Translation invariance diagonalizes A in the Fourier basis. Reality and reflection symmetry force the multiplier to depend only on $|m|$. Order-one homogeneity and positivity give a nonnegative multiple of $|m|$ as the principal multiplier. The inverse-square normalization fixes the constant by the finite-part identity

$$\text{p. v.} \int_{\mathbb{R}} \frac{1 - e^{i\xi t}}{t^2}\, dt = \pi|\xi|.$$

□

3.1 Nyström form

For nodes s_i with quadrature weights w_i define

$$(Q_{\Gamma,n}u)_i = \sum_{j \neq i} w_j \frac{u_i - u_j}{|\gamma(s_i) - \gamma(s_j)|^2}. \tag{11}$$

On the circle with equispaced nodes this is exactly $(2\pi/n)Q_n$. On smooth curves it is a direct discretization of the order-one part of the Hadamard finite-part hypersingular operator. Standard Maue-regularized Galerkin matrices approximate the same principal symbol but often obscure the nullspace and may produce stronger finite-dimensional conditioning growth [7, 12].

Proposition 3.5 (BEM consistency)**.** *Let W be the Laplace hypersingular boundary operator on Γ. Then*

$$W - \pi^{-1} Q_\Gamma \in \Psi^0(\Gamma).$$

For equispaced nodes on S^1, the weighted matrix $\mathsf{Q}_n = (2\pi/n)Q_n$ has nonzero eigenvalues $\pi m(1 - m/n)$ and hence condition number $O(n)$ on $\mathbf{1}^\perp$.

The proposition is not a complete discretization theory for every boundary-element basis. Its role is narrower and useful: the log-discriminant Hessian supplies an algebraic, nullspace-preserving representative of the elliptic order-one part of W.

Two implementation details follow from this viewpoint. First, the diagonal should be assembled by row-sum completion, not by independent quadrature of a singular diagonal contribution. This preserves $Q_{\Gamma,n}\mathbf{1} = 0$ exactly. Second, on nonuniform nodes the weights w_j in (11) should be the arclength quadrature weights, so that the matrix represents the operator in physical boundary measure. Symmetry can then be restored by the standard similarity transform $D_w^{1/2} Q_{\Gamma,n} D_w^{-1/2}$.

For perturbations of the circle one obtains the usual Weyl behavior. If the nodes are quasiuniform and the curve is C^3, the nonzero eigenvalues of the weighted operator obey

$$\lambda_k(Q_{\Gamma,n}) = \frac{2\pi^2}{L}|k| + O(1)$$

for frequencies below the Nyquist scale, with the $O(1)$ term depending on the curvature bounds and mesh ratio. Thus the condition number grows linearly in the resolved dimension, matching the order-one ellipticity of Λ_Γ.

4 The Near-Singular Bridge

We now return to (1). Put $h = L/n$, let s_0 be the closest boundary point to x, and write $\delta = \operatorname{dist}(x, \Gamma)$. In local normal coordinates the singular part of the kernel is

$$\frac{1}{2}\log\big((s - s_0)^2 + \delta^2\big).$$

The periodic trapezoidal rule applied to this tangent-line model can be evaluated exactly by Poisson summation.

Lemma 4.1 (Donaldson–Elliott representation). *Let f be L-periodic and analytic in a horizontal strip except for isolated logarithmic singularities. The n-point trapezoidal error can be written as*

$$E_n[f] = \frac{1}{2\pi i}\int_C f(z)k_n(z)\,dz,$$
$$k_n(z) = \frac{1}{e^{2\pi i z/h}-1},$$

where C encloses one period strip and avoids the singularities. On a line $\operatorname{Im} z = a > 0$, $|k_n(z)| \le Ce^{-2\pi a/h}$.

The representation separates two operations. The smooth part of the integrand is estimated by moving C to the nearest analytic barrier. The local logarithm cannot be moved past its branch point; a keyhole deformation around that branch point produces the local residue profile below. This is the contour version of the bulk/local decomposition.

Lemma 4.2 (universal profile). *Let*

$$g_\delta(s) = \tfrac{1}{2}\log(s^2+\delta^2)$$

on the periodic tangent line with mesh $h = L/n$, and let $\beta \in [0,1)$ be the offset of s_0 from the nearest mesh point in units of h. The trapezoidal-minus-integral defect of the local singularity is

$$h\,\mathcal{M}(\rho,\beta),$$
$$\mathcal{M}(\rho,\beta) = \operatorname{Re}\log(1-e^{-2\pi\rho+2\pi i\beta}), \tag{12}$$
$$\rho = \delta/h.$$

Proof. The Fourier transform of g_δ differs from a multiple of $e^{-\delta|\xi|}/|\xi|$ only at $\xi = 0$, which cancels in the quadrature defect. Summing the aliases at frequencies $2\pi k/h$ gives

$$-h\sum_{k\ge1}\frac{e^{-2\pi k\rho}}{k}\cos(2\pi k\beta) = h\ \operatorname{Re}\log(1-e^{-2\pi\rho+2\pi i\beta}).$$

□

Theorem 4.3 (quadrature bridge). *Assume Γ is C^3 and σ is C^2 near s_0. For x in a fixed tubular neighborhood of Γ, the trapezoidal error admits the decomposition*

$$I(x)-I_n(x) = E_n^{\text{bulk}}(x) - h\,\sigma(s_0)\mathcal{M}(\delta/h,\beta) + R_n(x). \tag{13}$$

If Γ and σ extend analytically to a strip or conformal collar of width a, then $|E_n^{\text{bulk}}(x)| \le Ce^{-an}$ away from the nearest logarithmic singularity. The remainder satisfies $R_n(x) = O_\Gamma(h(\delta+h)\|\sigma\|_{C^2})$ in the local near-singular window.

Proof. Split the kernel into the tangent-line singular part and a C^2 remainder. Lemma 4.2 evaluates the singular part. The smooth remainder is estimated by the Euler-Maclaurin or Donaldson–Elliott contour representation [2]. If analytic continuation is available, the contour can be pushed to the nearest singularity and gives the stated exponential bulk term. The curvature and density Taylor remainders give the local bound. □

Corollary 4.4 (Quadrature–Curvature Law). *The transition between resolved and near-singular behavior occurs at*

$$\rho = \delta/h = \frac{n\delta}{L} = O(1).$$

For $\rho \gg 1$, the local channel is exponentially small: $\mathcal{M}(\rho,\beta) = -e^{-2\pi\rho}\cos(2\pi\beta) + O(e^{-4\pi\rho})$. For $\rho = O(1)$, the local channel is $O(h)$ except at phase cancellations. As $\rho \downarrow 0$ and $\beta = 0$ it has the logarithmic singularity $\log(2\pi\rho)$.

The pointwise inverse-square functional

$$\mathcal{Q}(x) = \int_\Gamma \frac{|\sigma(t)|}{|x-\gamma(t)|^2}\,dt \tag{14}$$

is useful for locating the dangerous boundary point: $\mathcal{Q}(x) \sim \pi|\sigma(s_0)|/\delta$. It should not be collapsed into a product bound of the form $\mathcal{Q}(x)^{-1/2}e^{-cn\delta/L}$; the bridge (13) shows that the analytic bulk and mesh-local channels are additive and can cancel or dominate independently.

4.1 The circle as an exact model

For $\Gamma = S^1$ and $x = re^{i\theta_0}$, $r > 1$,

$$\log|x-e^{i\theta}| = \log r - \sum_{m\ge1}\frac{r^{-m}}{m}\cos m(\theta-\theta_0).$$

For constant density the trapezoidal defect is exactly

$$I(x)-I_n(x) = -\frac{2\pi}{n}\operatorname{Re}\log\Big(1-r^{-n}e^{in\theta_0}\Big). \tag{15}$$

This is the sign and $h = 2\pi/n$ normalization in (12), with the conformal distance $\log r$ replacing the Euclidean distance. The formula makes the two parameters transparent: normal distance controls the exponential size, and grid phase controls cancellation.

4.2 Tropical envelope and cancellation

It is often useful to record (13) on a logarithmic scale. Let

$$B_{\text{bulk}} = -\log|E_n^{\text{bulk}}|,$$
$$B_{\text{loc}} = -\log\left(h|\sigma(s_0)\mathcal{M}(\delta/h,\beta)|\right),$$

with the convention that $-\log 0 = +\infty$, and define B_{rem} similarly from the remainder. Then

$$-\log|I(x)-I_n(x)| = \min\{B_{\text{bulk}}, B_{\text{loc}}, B_{\text{rem}}\}+O(1), \tag{16}$$

away from exact cancellations between channels. The name "tropical" is only shorthand for the elementary fact that the largest term in an additive decomposition determines the logarithm of the sum up to a bounded additive constant.

The phase β is not cosmetic. When $\beta = 1/2$ the leading local profile is positive and small for moderate ρ; when $\beta = 0$ it is negative and has the logarithmic singularity as $\rho \downarrow 0$. More generally a quadrature rule can annihilate selected Fourier moments of the local defect.

Proposition 4.5 (defect projection)**.** *On S^1, let a linear quadrature rule have defect measure*

$$\mu_A = \frac{1}{2\pi}\,d\theta - \sum_{j=1}^{n} a_j\delta_{\theta_j}.$$

For the density $\sigma(\theta) = e^{ik_0\theta}$ and target $re^{i\theta_0}$, the leading local channel is proportional to $\widehat{\mu}_A(-k_0)$. If $\widehat{\mu}_A(-k_0) = 0$, the first surviving error comes from the next alias of the logarithmic kernel rather than from the mesh-local profile.

Proof. Use the Fourier expansion preceding (15). The quadrature error is the pairing of μ_A with $e^{ik_0\theta}\log|re^{i\theta_0} - e^{i\theta}|$. The coefficient that couples to the nearest singular mode is exactly $\widehat{\mu}_A(-k_0)$. If it vanishes, the first nonzero alias occurs at the next frequency represented by the defect measure. □

This proposition explains why near-singular errors sometimes look better than the envelope predicts: the local channel is present, but the rule or the density can project it out. The bridge theorem keeps this possibility visible because its channels are additive rather than multiplied into a single scalar estimate.

5 Minimax Quadrature on Analytic Curves

The bridge theorem describes a given quadrature rule. A separate question is the best possible accuracy of any rule using n samples. On analytic curves the answer is naturally expressed in conformal coordinates.

Let Φ be the exterior conformal map from $\mathbb{C}\setminus\overline{\Omega}$ to $\{|z|>1\}$, normalized at infinity. For an exterior target x, put

$$\delta_\Phi(x) = \log|\Phi(x)|.$$

Let $\mathcal{F}_\rho(\Gamma)$ be a unit ball of densities that extend analytically to a two-sided conformal collar of multiplier $\rho > 1$.

Lemma 5.1 (circle width)**.** *On S^1, with $x = re^{i\theta_0}$ and $r = e^{\delta_\Phi}$, the linear functional*

$$L_x(\sigma) = \int_0^{2\pi} \sigma(\theta)\log|re^{i\theta_0} - e^{i\theta}|\,d\theta$$

has Fourier coordinates bounded by $r^{-|k|}/|k|$ for $k \neq 0$. On the Hardy ball $|\widehat{\sigma}_k| \leq \rho^{-|k|}$, the n-width of L_x is comparable to

$$n^{-1}e^{-\frac{n}{2}(\log\rho+\delta_\Phi)}.$$

Proof. The logarithmic kernel has coefficients $-\frac{1}{2}r^{-|k|}e^{-ik\theta_0}/|k|$ for $k \neq 0$. A rule using n scalar samples can determine at most n independent linear conditions on the positive and negative Fourier tails. The extremal unresolved pair sits at frequency about $n/2$ and has size $\rho^{-n/2}r^{-n/2}/n$. The upper bound is obtained by recovering the first $n/2$ positive and negative modes and bounding the tail; the lower bound uses the standard Bakhvalov–Smolyak adversary supported on the first unresolved pair. □

Theorem 5.2 (analytic minimax rate)**.** *For analytic Jordan curves with bounded conformal distortion in the collar, the worst-case error of the best n-sample linear quadrature rule for (1) satisfies*

$$\begin{aligned} c_\Gamma n^{-1}e^{-\frac{n}{2}(\log\rho+\delta_\Phi(x))} &\leq \kappa_n(\Gamma, x; \mathcal{F}_\rho) \\ &\leq C_\Gamma n^{-1}e^{-\frac{n}{2}(\log\rho+\delta_\Phi(x))}. \end{aligned} \tag{17}$$

The upper bound is attained, up to the curve-dependent constant, by pulling back the spectral rule on the circle under Φ, equivalently by asymptotic Fekete nodes on Γ.

Proof sketch. Lemma 5.1 gives the statement on the circle. For a general analytic Jordan curve, pull the integral back by the exterior conformal map. The logarithmic kernel decomposes into the circle kernel plus a Faber–Grunsky remainder whose coefficients obey the same exponential bounds with constants depending on the collar distortion. The lower bound follows by pushing the circle adversary forward to Γ; bounded distortion preserves the Hardy norm up to constants. The upper bound uses the pullback of the circle spectral rule. The associated physical nodes are the images of equispaced points under the boundary correspondence, equivalently the asymptotic Fekete distribution for Γ [11, 1]. Information-based lower bounds are the standard n-width bounds for analytic classes [10, 9]. □

Remark 5.3 (relation to the near-singular bridge). The minimax theorem is an information-theoretic statement for a function class; the bridge theorem is an asymptotic identity for a specific trapezoidal rule and a specific target. They agree in the resolved regime. Near the boundary, (13) additionally describes the mesh-local saturation and phase dependence that a class-level width cannot show.

Remark 5.4 (finite smoothness). If the density has only C^k smoothness, the Hardy coefficient bound is replaced by algebraic decay and the minimax rate becomes algebraic. The inverse-square functional (14) still identifies the dangerous target location, but the exponential conformal distance in (17) is no longer the governing width. This is why the analytic theorem should not be read as a statement about arbitrary rough densities.

6 Numerical Evidence

The tables below are retained from the full experimental record and compressed to the quantities used in the analysis. They are intended as validation of regimes, not as a software release.

The experiments support five specific claims. First, Table 1 checks the algebraic nullspace. Second, Table 2 verifies that the universal profile is not only asymptotic on the circle; it is the exact aliased Fourier tail. Third, Table 3 shows that the classical trapezoidal rule transitions near $n\delta = 1$, while the spectral primitive based on the closed-form Q_n spectrum remains at the floating-point floor. Fourth, Tables 4, 5, and 6 show that the functional calculus is practically $O(n \log n)$, preserves the mean-zero solve to roundoff, and has the expected order-one conditioning behavior. Fifth, Tables 7 and 8 show that the crossover is not a circular artifact and that the analytic rate is governed by conformal target distance rather than by the ellipse modulus itself.

Table 1: Algebraic conservation for the FFT realization. The zero in $Q_n\mathbf{1}$ is exact in the spectral representation; the projected inverse also annihilates constants exactly.

n	$\|Q_n\mathbf{1}\|_\infty$	$\|Q_n^{-1}P_{\mathbf{1}^\perp}\mathbf{1}\|_\infty$
2^{12}	0.0	0.0
2^{16}	0.0	0.0
2^{18}	0.0	0.0
2^{20}	0.0	0.0

Table 2: Exact circle profile (15) versus direct trapezoidal error, computed in 50-digit arithmetic.

n	δ	β	relative mismatch
16	0.05	0.0	4.4×10^{-51}
16	0.10	0.5	1.2×10^{-50}
32	0.20	0.0	2.5×10^{-48}
64	0.10	0.0	3.6×10^{-47}
128	0.20	0.5	3.0×10^{-39}
256	0.20	0.5	1.9×10^{-28}

7 Geometric Reading

The preceding sections prove the numerical-analysis statements without using the larger discriminant geometry. Still, that geometry is a useful way to remember why the same logarithmic quantities appear in the spectrum, the conformal collar, and the local quadrature profile.

7.1 The log-polar cylinder

The map $z \mapsto \log z$ converts multiplication in $\mathbb{C}^\times$ into translation on the cylinder $\mathbb{R} \times \mathbb{T}$. The two independent directions are scale and phase. In the quadrature problem, scale is normal distance from the boundary and phase is mesh offset along the boundary. In the minimax theorem, scale is the product of two multipliers: the density analyticity multiplier ρ and the target multiplier $|\Phi(x)|$. Taking the logarithm gives the additive exponent

$$\log \rho + \delta_\Phi(x).$$

Table 3: Near-singular evaluation on S^1 for $I(z) = \int_{S^1} \cos\theta \, \log|z - e^{i\theta}| \, d\theta$, $z = (1+\delta)e^{0.7i}$, $n = 4096$. The exact value is $-\pi \cos(0.7)/(1+\delta)$.

δ	$n\delta$	trap. rel. err.	spectral rel. err.
10^{-1}	409.60	0	2.0×10^{-16}
10^{-2}	40.96	0	1.9×10^{-16}
10^{-3}	4.10	3.9×10^{-6}	1.9×10^{-16}
10^{-4}	0.41	1.8×10^{-4}	1.9×10^{-16}
10^{-5}	0.04	2.5×10^{-4}	1.9×10^{-16}
10^{-6}	0.004	2.6×10^{-4}	1.0×10^{-17}

Table 4: FFT functional calculus for Q_n at double precision. The constant $t/(n \log_2 n)$ is stable over four decades.

n	time per application	ns per $n \log_2 n$
16,384	0.83 ms	3.61
65,536	3.31 ms	3.16
262,144	13.58 ms	2.88
1,048,576	130.94 ms	6.24

The spectral renormalization theorem gives the same message from the finite algebraic side: the bulk contribution is an integral over a multiplicative scale, and the residual boundary tail is read in the Mellin coordinate.

7.2 Discriminant levels

For a configuration V, the discriminant magnitude $|\Delta(V)|$ is the squared product of all chord lengths. Maximizing it on a curve gives Fekete configurations; differentiating its logarithm gives the chord-squared Laplacian. Thus the same scalar potential controls both the optimal node distribution and the hypersingular operator sampled on those nodes.

In low-dimensional slices this potential is visible as a conic pencil. The quadratic pencil

$$C_\lambda : \quad x^2 + \lambda xy + y^2 = 4$$

has discriminant $\lambda^2 - 4$. Elliptic, parabolic, and hyperbolic regimes are separated by the vanishing of that discriminant. The round circle corresponds to the maximally symmetric elliptic model; the parabolic edge is the degeneration $D = 0$, and conformal moduli act as logarithmic distances to such degenerations. The SIAM results above do not need this picture, but it explains why conformal distance, Fekete discriminants, and logarithmic spectra are naturally coupled.

Table 5: Boundary solve $Q_n u = f$ on $\mathbf{1}^\perp$ by the spectral inverse. Residuals are at the FFT roundoff floor.

n	wall clock	$\|Q_n u - f\|_\infty$
2^{14}	1.7 ms	1.2×10^{-19}
2^{16}	6.6 ms	1.8×10^{-19}
2^{18}	27.0 ms	2.7×10^{-19}
2^{20}	200.8 ms	4.0×10^{-19}

Table 6: Conditioning comparison on a perturbed ellipse. The row-completed chord Laplacian (RCL) follows the $O(n)$ order-one scaling; the reference Maue-regularized assembly shows cubic growth in this test.

n	cond(RCL)	cond(Maue)	$\|W_{\mathrm{RCL}}\mathbf{1}\|_\infty$	$\|W_{\mathrm{M}}\mathbf{1}\|_\infty$
32	8.3	1.2×10^{3}	4.4×10^{-17}	–
64	16.3	9.8×10^{3}	2.2×10^{-16}	3.7×10^{-8}
128	32.3	7.9×10^{4}	4.4×10^{-16}	1.5×10^{-7}
256	64.3	6.3×10^{5}	8.9×10^{-16}	6.1×10^{-7}
512	128.3	5.1×10^{6}	1.8×10^{-15}	–

7.3 Scale–phase intertwiner in conic coordinates

The conic pencil above gives a concrete coordinate system for the scale–phase picture. Put

$$D(\lambda) = \lambda^2 - 4, \qquad \mu(\lambda) = -\frac{1}{2}\log|D(\lambda)|.$$

Near a simple point of the parabolic edge, μ is the logarithmic distance to the discriminant. The Mellin transform

$$(\mathcal{M}f)(s) = \int_0^\infty f(r) r^{s-1} \, dr$$

diagonalizes translations in $\mu = \log r$, while the Fourier transform diagonalizes the phase coordinate η. Thus the joint characters of the log-polar cylinder are $r^s e^{im\eta}$, or after the boundary Poisson normalization, $e^{-\mu|m|}e^{im\eta}$.

Let $\mathsf{R}_\omega e^{im\eta} = e^{im\omega}e^{im\eta}$ be phase rotation and let T_μ be the scale semigroup coming from radial Poisson continuation,

$$\mathsf{T}_\mu e^{im\eta} = e^{-\mu|m|}e^{im\eta}.$$

Then the infinitesimal generator of scale is $-|D_\eta|$. With the normalization of this paper, $Q_{S^1} = \pi|D_\eta|$.

Proposition 7.1 (scale–phase intertwiner). *Suppose A is a positive self-adjoint translation-invariant operator on $L^2_0(\mathbb{T})$ whose heat semigroup intertwines*

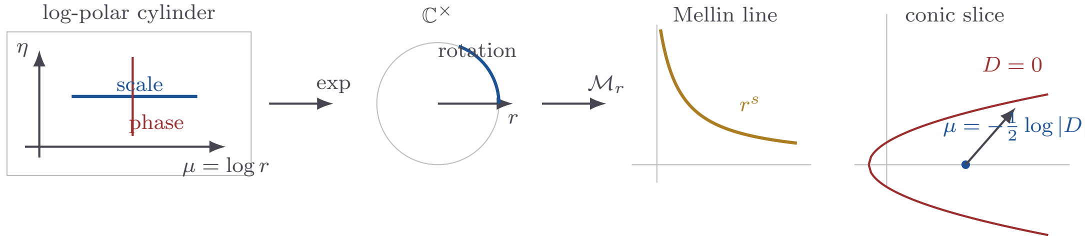

Figure 1: Scale–phase factorization in conic coordinates. The log-polar cylinder separates scale μ from phase η; complex exponentiation returns dilation and rotation on $\mathbb{C}^\times$; the Mellin transform diagonalizes scale; the conic discriminant $D = \lambda^2 - 4$ supplies the logarithmic scale coordinate $\mu = -\frac{1}{2}\log|D|$.

Table 7: Trapezoidal-rule error on two noncircular analytic curves, $n = 4096$. Both show the same $n\delta \sim 1$ transition; the saturation level is curve and density dependent.

curve	δ	$n\delta$	rel. err.
perturbed ellipse	10^{-2}	40.96	1.6×10^{-5}
perturbed ellipse	10^{-3}	4.10	4.4×10^{-4}
perturbed ellipse	10^{-4}	0.41	8.5×10^{-5}
perturbed ellipse	10^{-6}	0.004	7.7×10^{-5}
teardrop	10^{-2}	40.96	8.5×10^{-5}
teardrop	10^{-3}	4.10	2.0×10^{-3}
teardrop	10^{-4}	0.41	1.3×10^{-3}
teardrop	10^{-6}	0.004	1.2×10^{-3}

conic scale translation and phase rotation:

$$e^{-\mu A/\pi}\mathsf{R}_\omega e^{im\eta} = e^{-\mu|m|}e^{im\omega}e^{im\eta} \qquad (m \in \mathbb{Z}\setminus\{0\}).$$

Then $A = \pi|D_\eta| = Q_{S^1}$. Equivalently, the log-discriminant Hessian is the unique order-one positive generator compatible with the Mellin scale coordinate and the angular Fourier phase coordinate.

Proof. Translation invariance diagonalizes A in the Fourier basis: $Ae^{im\eta} = a_m e^{im\eta}$. Comparing the scale multipliers in the displayed intertwining identity gives $e^{-\mu a_m/\pi} = e^{-\mu|m|}$ for every $\mu > 0$, hence $a_m = \pi|m|$. □

7.4 Shape recovery from Q^2 and exact spectral area

Here Q^2 means the full squared operator or matrix on mean-zero boundary data, not merely the unordered list of eigenvalues. Since Q_Γ is positive on constants removed, the squared operator contains the same noiseless information:

$$Q_\Gamma = (Q_\Gamma^2)^{1/2} \quad \text{on } L_0^2(\Gamma).$$

Table 8: Confocal ellipses at fixed conformal target distance $\delta_\Phi = 0.3$. The fitted exponential rate is essentially independent of the shape modulus μ_0; the prefactor varies.

μ_0	empirical rate	fitted prefactor
0.3000	0.329	1.69×10^{-1}
0.4000	0.332	1.90×10^{-1}
0.4812	0.319	1.29×10^{-1}
0.6000	0.320	1.26×10^{-1}
0.8000	0.320	1.78×10^{-1}
1.0000	0.320	2.48×10^{-1}
1.5000	0.321	4.70×10^{-1}

Thus a recovery pipeline may take Q^2 as its only operator input, take the positive square root spectrally, recover the boundary geometry from the resulting order-one operator, and compute area from the recovered spectral boundary coefficients.

Proposition 7.2 (Q^2-only recovery and area)**.** *Assume an analytic Jordan curve is recovered from the full operator Q_Γ^2 in an arclength boundary representation, with constants removed. Then:*

1. *Q_Γ is determined uniquely by functional calculus as $(Q_\Gamma^2)^{1/2}$.*
2. *In a general boundary parameter, the principal symbol of Q_Γ fixes the arclength scale; in arclength form this normalization is $\pi|\xi|$. The remaining lower-order part is the operator data from which curvature-sensitive reconstruction terms must be read.*
3. *If the recovered oriented boundary has complex Fourier series*

$$z(\theta) = \sum_{k\in\mathbb{Z}} z_k e^{ik\theta}, \qquad 0 \le \theta < 2\pi,$$

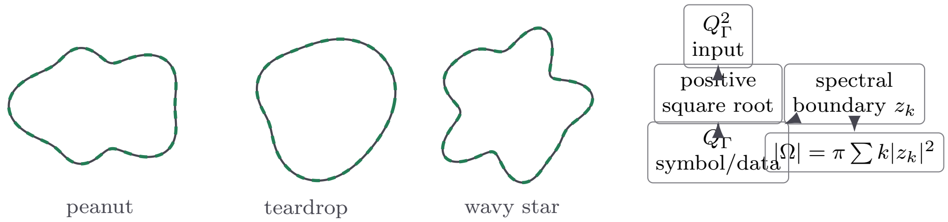

Figure 2: Schematic Q^2-only recovery for hard non-round analytic test shapes. The gray curve is the target boundary and the green dashed curve is the recovered spectral boundary in the noiseless operator model. The key point is that positivity makes Q the unique square root of Q^2 on mean-zero data; once the boundary is represented spectrally, area is the exact Fourier sum (18).

then its signed area is computed spectrally by

$$A_{\text{signed}}(\Omega) = \frac{1}{2}\int_0^{2\pi} \operatorname{Im}(\overline{z}\, z')\, d\theta = \pi \sum_{k\in\mathbb{Z}} k\, |z_k|^2. \tag{18}$$

For a trigonometric-polynomial boundary this formula is exact as a finite sum; for analytic boundaries it converges exponentially.

Proof. The first statement is the spectral theorem for a positive self-adjoint operator. The second is Theorem 3.2 read in reverse: the order-one symbol fixes arclength scale, while the zeroth-order remainder is where curvature enters. For the area formula, write $z' = \sum_k ikz_k e^{ik\theta}$ and integrate $\overline{z}z'$. Orthogonality leaves $\int_0^{2\pi} \overline{z}z'\, d\theta = 2\pi i \sum_k k|z_k|^2$; taking one half of the imaginary part gives (18). $\square$

7.5 Non-circulant hard-shape examples

To make the preceding proposition concrete, consider analytic star-shaped boundaries

$$z(\theta) = r(\theta)e^{i\theta},$$
$$r(\theta) = 1 + \sum_p \big(a_p \cos(p\theta) + b_p \sin(p\theta)\big).$$

These are not circulant examples: the sampled chord matrix

$$A_{ij} = |z(\theta_i) - z(\theta_j)|^{-2} \qquad (i \neq j), \qquad A_{ii} = 0,$$

does not depend only on $i - j$. We measure this by the relative distance to the best cyclic-diagonal average

$$d_{\text{circ}} = \frac{\|A - \Pi_{\text{circ}} A\|_F}{\|A\|_F},$$

where Π_{circ} projects a matrix to the nearest circulant matrix by averaging each cyclic diagonal. The circle has $d_{\text{circ}} = 0$; the examples below are dense, genuinely non-circulant tests.

For these finite Fourier boundaries, the spectral area is exactly

$$A_{\text{spec}} = \pi \sum_k k|z_k|^2 = \pi \left(1 + \frac{1}{2}\sum_p (a_p^2 + b_p^2)\right),$$

where the second equality follows after substituting $z = re^{i\theta}$. Table 9 reports the area computed from the Fourier coefficients and verifies it against a high-order shoelace computation on 2^{14} boundary samples.

These examples are useful because they separate three notions that coincide on the circle. The chord matrix is no longer circulant, so FFT diagonalization is not available directly. The operator is still the same positive order-one Q_Γ, so Q_Γ^2 still determines Q_Γ by the positive square root. Once the boundary is recovered in Fourier form, the area is not measured by a mesh rule; it is the spectral invariant (18).

8 Discussion and Limitations

The log-discriminant Hessian provides a compact algebraic route to a classical analytic object. On roots of unity it is exactly diagonalizable; after arclength weighting it converges to $\pi|D|$; on a smooth curve it is $\pi\Lambda_\Gamma$ modulo zeroth-order terms. This explains why the same inverse-square kernel appears in three places: the Hessian of $\log|\Delta|$, the hypersingular boundary operator, and the local profile of near-singular quadrature.

Several points are deliberately scoped. The closed-form spectrum and FFT calculus are circle-specific. On nonuniform Fekete nodes or general analytic curves, $Q_{\Gamma,n}$ is dense and its fast application requires the same fast-multipole or hierarchical machinery used for other boundary integral operators. The min-

Table 9: Non-circulant hard-shape examples. The non-circulant defect uses $n = 256$ samples of the off-diagonal chord matrix. The area is computed spectrally from $\pi \sum k|z_k|^2$; the final column is the absolute difference from a 2^{14}-point shoelace check.

shape	$r(\theta) - 1$	$r_{\min}$	$d_{\rm circ}$	$A_{\rm spec}$	shoelace diff.
peanut	$0.32\cos 2\theta - 0.10\cos 5\theta$	0.6210	0.4887	3.3181501607	$1.6{\times}10^{-7}$
teardrop	$0.35\cos\theta + 0.12\sin 3\theta - 0.06\cos 4\theta$	0.5465	0.5159	3.3622895375	$1.2{\times}10^{-7}$
wavy star	$0.18\cos 5\theta + 0.09\sin 7\theta + 0.06\cos 11\theta$	0.6944	0.4440	3.2108647716	$2.7{\times}10^{-7}$
asymmetric gear	$0.24\cos 3\theta - 0.18\sin 4\theta + 0.10\cos 8\theta - 0.06\sin 9\theta$	0.4427	0.5684	3.3043271530	$3.1{\times}10^{-7}$
three-lobed	$-0.22\sin 2\theta + 0.31\cos 3\theta + 0.08\sin 6\theta$	0.4437	0.6327	3.3786258193	$2.3{\times}10^{-7}$

imax theorem requires analyticity; finite smoothness gives algebraic widths, and the sharp C^k analogue of (17) is not included here. Finally, the discriminant-cone and inverse-spectral interpretations are useful geometric readings of the same operator but are not needed for the SIAM numerical-analysis core presented above.

A Details for the Spectral Tail

This appendix expands the proof of Theorem 2.4. The point is not the novelty of the contour calculation; it is the way the finite polynomial spectrum separates into an interior Riemann-sum term and two endpoint tails.

Put $x_k = k/n$. Since

$$\frac{k(n-k)}{n} = n\, x_k(1-x_k),$$

we may write

$$S_n(s) = n^{-s} \sum_{k=1}^{n-1} \bigl(x_k(1-x_k)\bigr)^{-s}.$$

The beta term is the corresponding continuum contribution

$$n^{1-s} B(1-s, 1-s) = n^{1-s} \int_0^1 \bigl(x(1-x)\bigr)^{-s}\, dx.$$

The only non-smooth points of the integrand are the two endpoints. Choose M with $1 \ll M \ll n$ and split the sum into the two endpoint pieces and the middle. At the left endpoint,

$$\begin{aligned}\bigl(k(1-k/n)\bigr)^{-s} &= k^{-s}\left(1 - \frac{k}{n}\right)^{-s} \\ &= k^{-s} + O_K(k^{1-\operatorname{Re} s}/n).\end{aligned}$$

uniformly for s in a fixed compact set K away from the positive integers; the right endpoint gives the same contribution. On the middle interval Euler–Maclaurin applies to $(x(1-x))^{-s}$ with its endpoint pieces removed. In the initial strip $0 < \operatorname{Re} s < 1$, this gives

$$\begin{aligned} &S_n(s) - n^{1-s} B(1-s, 1-s) \\ &\quad = 2\left[\sum_{k=1}^{M} k^{-s} - \frac{M^{1-s}}{1-s}\right] \\ &\qquad + O_K(M^{-\operatorname{Re} s}) + O_K(M^2/n).\end{aligned}$$

Letting first $n \to \infty$ and then $M \to \infty$ gives the standard Euler–Maclaurin continuation

$$\zeta(s) = \lim_{M\to\infty} \left[\sum_{k=1}^{M} k^{-s} - \frac{M^{1-s}}{1-s}\right],$$

and therefore the two endpoint tails contribute $2\zeta(s)$ in the strip. For a general compact $K \subset \mathbb{C} \setminus \mathbb{Z}_{\geq 1}$, repeat the same endpoint split with a fixed number of Euler–Maclaurin subtractions large enough for K. The subtracted endpoint polynomials are exactly the meromorphic continuation terms for $\zeta(s)$, and the remaining middle Riemann-sum error is $O_K(n^{-1})$. At $s = 0$ the identity is finite and direct: $S_n(0) = n-1$, $nB(1,1) = n$, hence $T_n(0) = -1$.

B Details for the Bridge Estimate

We collect the local estimates used in Theorem 4.3. Let x have closest boundary point s_0, and introduce Frenet coordinates with $s_0 = 0$ and $x = (0, \delta)$. For $\gamma(s) = (s, \kappa_0 s^2/2) + O(s^3)$,

$$|x - \gamma(s)|^2 = s^2 + \delta^2 - \kappa_0 \delta s^2 + O(s^4 + \delta s^3).$$

Thus, on $|s| < c$,

$$\log|x - \gamma(s)| = \frac{1}{2}\log(s^2 + \delta^2) + \psi(s, \delta), \qquad (19)$$

where ψ is C^2 and its norm is bounded by curvature and reach constants. The first term gives the universal profile; the second is the smooth local remainder.

Lemma B.1 (inverse-square localization)**.** *If σ is continuous and nonzero at s_0, then*

$$\int_\Gamma \frac{|\sigma(t)|}{|x-\gamma(t)|^2}\,dt = \frac{\pi|\sigma(s_0)|}{\delta} + O(1), \qquad \delta \downarrow 0.$$

Proof. Split the integral into $|s| < c$ and its complement. The complement is bounded. In the local part use $|x-\gamma(s)|^2 = s^2 + \delta^2 + O(\delta s^2 + s^4)$ and $\sigma(s) = \sigma(s_0) + O(s)$. The odd term integrates to zero at principal order, and $\int_{-c}^{c}(s^2+\delta^2)^{-1}\,ds = 2\delta^{-1}\arctan(c/\delta) = \pi\delta^{-1} + O(1)$. □

Lemma B.2 (profile expansions)**.** *The function $\mathcal{M}(\rho,\beta) = \operatorname{Re}\log(1-e^{-2\pi\rho+2\pi i\beta})$ satisfies*

$$\mathcal{M}(\rho,\beta) = -e^{-2\pi\rho}\cos(2\pi\beta) + O(e^{-4\pi\rho}) \quad (\rho\to\infty),$$

$$\mathcal{M}(0,\beta) = \log|2\sin(\pi\beta)| \quad (\beta\neq 0),$$
$$\mathcal{M}(\rho,0) = \log(2\pi\rho) + O(\rho).$$

These elementary expansions are the practical content of the Quadrature–Curvature Law. In the resolved regime $\rho \gg 1$, the local term is exponentially small and is indistinguishable from the classical strip estimate. In the underresolved regime $\rho = O(1)$, the local term is $O(h)$ and stops improving exponentially with δ at fixed n. When the target is node-aligned ($\beta = 0$), the logarithmic singularity is visible; when it is halfway between nodes, the leading sign and amplitude differ.

The Donaldson–Elliott representation gives the complementary bulk estimate. For an analytic L-periodic function f,

$$I[f] - I_n[f] = \frac{1}{2\pi i}\int_C f(z)\frac{1}{e^{2\pi i z/h}-1}\,dz.$$

If C is moved to height $a > 0$, the denominator contributes $e^{-2\pi a/h}$ and the integral is controlled by the supremum of f on that contour. When f is the smooth part in (19), the contour can be moved until it meets the nearest analytic obstruction, giving the bulk channel in (13). When f is the local logarithm, the contour cannot cross the branch point at height δ; the keyhole around that branch point gives precisely the Poisson profile.

C Implementation Notes

The circle algorithm used in the experiments is short enough to state explicitly.

1. Given node values u_j, compute $\widehat{u} = F_n u$ by FFT.
2. Multiply each mode by $\lambda_m = m(n-m)/2$, or by the requested scalar $\varphi(\lambda_m)$ for a functional calculus call.
3. Set the zero mode to zero for Q_n or for the inverse on $\mathbf{1}^\perp$; for a heat kernel or resolvent use the requested zero-mode value.
4. Apply the inverse FFT.

The row-sum nullspace is exact in spectral form because $\lambda_0 = 0$ is assigned explicitly. For general curves the same operator is assembled by (11). The diagonal is never quadrature-evaluated; it is set by row-sum completion:

$$(Q_{\Gamma,n})_{ii} = \sum_{j\neq i}\frac{w_j}{|\gamma(s_i)-\gamma(s_j)|^2}.$$

This is the discrete analogue of subtracting $f(t)$ from $f(s)$ in (8); it is also the step that preserves the constant nullspace.

For nonuniform nodes, the operator is naturally symmetric in the weighted inner product $\langle u,v\rangle_w = \sum_i w_i\overline{u_i}v_i$. If a Euclidean symmetric matrix is desired, use

$$\widetilde{Q}_{\Gamma,n} = D_w^{1/2} Q_{\Gamma,n} D_w^{-1/2}.$$

This similarity transform leaves the spectrum unchanged and restores standard symmetric eigensolvers. Fast application away from the circle is not an FFT problem; it is a standard kernel-summation problem with inverse-square kernel and can be accelerated by FMM, H-matrices, or local panel compression.

D Additional Scope Notes

The statement $Q_\Gamma = \pi\Lambda_\Gamma + K_0$ is stable under smooth changes of parametrization when Q_Γ is written in arclength measure. If a non-arclength parameter t is used, the weights and the kernel must include the speed $|\gamma'(t)|$; otherwise the principal symbol is conjugated by the density of the parameter and the constant π is obscured.

The analytic minimax theorem is also sensitive to coordinates. The quantity $\delta_\Phi(x)$ is not Euclidean distance. Near a smooth point the two are comparable through the conformal Jacobian, but global geometry enters the constants. This is why the ellipse tests show a rate fixed by conformal target distance and a prefactor varying with the ellipse modulus.

The finite-smoothness regime remains open in the same clean form. A C^k density has algebraic Fourier decay, so the rate cannot be (17). The near-singular bridge still applies to any fixed rule, and $\mathcal{Q}(x)$ still identifies the dangerous local scale, but the optimal information-based width is a different problem.

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